

22PAGMO3

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Problem

Let ABC be an acute triangle with $AB < AC$. Denote by P and Q points on the segment BC such that $\angle BAP = \angle CAQ < \frac{\angle BAC}{2}$. B_1 is a point on segment AC . BB_1 intersects AP and AQ at P_1 and Q_1 , respectively. The angle bisectors of $\angle BAC$ and $\angle CBB_1$ intersect at M . If $PQ_1 \perp AC$ and $QP_1 \perp AB$, prove that AQ_1MPB is cyclic.

Solution.

$$\angle Q_1PP_1 = 90^\circ - \angle CAP = 90^\circ - \angle BAQ = \angle AQP_1 = \angle Q_1QP_1$$

So P_1Q_1QP is a cyclic quadrilateral. Let $R = P_1Q \cap PQ_1$.

We know that the circumcenter of P_1Q_1QP (say O) is the orthocenter of $\triangle P_1AB$. Recall that $QP_1 \perp AB$. As a result O lies on P_1Q which makes P_1Q the diameter. As a result, $\angle AQ_1B = 90^\circ = \angle AP_1B$. So $ABPQ_1$ is cyclic. Obviously M is the midpoint of arc PQ_1 by Incenter-Excenter Lemma, we can conclude that AQ_1MPB is cyclic.